

Deep Learning Based Chatbots *are* Smarter

Contemporary chatbots extensively use machine learning, natural language processing, artificial intelligence and deep learning. They are typically used in the customer service space for almost all domains. Chatbots based on deep learning are far better than traditional variants. Here's why.

Chatbots are currently being used extensively to change customer behaviour. Usually, traditional artificial intelligence (AI) concepts are used in designing chatbots. However, modern applications generate such vast volumes of data that it becomes arduous to process this with traditional AI algorithms.

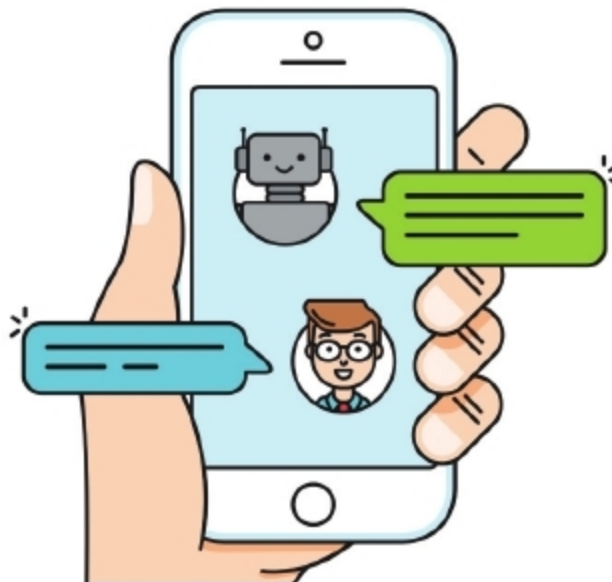
Deep learning is a subset of AI and is the most suitable technique to process large quantities of data. Deep learning based systems learn from copious data points. Systems like chatbots are the right contenders for deep learning as they require abundant data points to train the system to reach precise levels of performance. The main purpose of chatbots is to offer the most appropriate reply to any question or message that it receives. The ideal response from a chatbot has multiple aspects to it, such as:

- It should be able to chat in a pragmatic manner
- Respond to the caller's query
- Provide the corresponding, relevant information
- Raise follow up questions like in a real conversation

Deep learning simulates the human mind for processing information. It works like the human brain by categorising a variety of information, and automatically discovers the features to be used to classify this information in a way that is perfect for chatbot systems.

Steps for designing chatbots using deep learning

The goal while designing chatbots using deep learning is to entirely automate the system to lessen the need



for human management as much as possible. To achieve this, we need to completely replace all human experts with a chatbot, eradicating the need for client service representatives entirely. Figure 1 depicts the steps for designing chatbots using deep learning.

The first step when designing a chatbot is to collect the existing interactions between clients and service representatives, in order to teach the machine the phrases that are important while interacting with customers. This is called ontology creation.

The data preparation or data preprocessing is the next step in designing the chatbot. This consists of several steps such as tokenisation, stemming and lemmatising. This phase

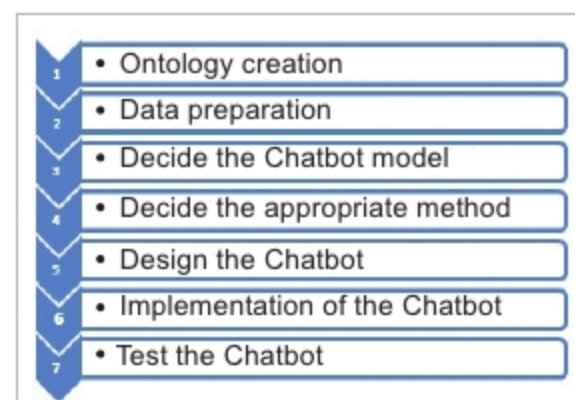


Figure 1: Steps for designing chatbots using deep learning

integrates grammar into machine understanding.

The third step involves deciding on the appropriate model of the chatbot. There are two prominent models — retrieval based and generative. Retrieval models apply the repository of predefined responses while generative models are advanced versions of the retrieval model that use deep learning concepts.

The next step is to decide on the appropriate technique to handle client interactions efficiently.

Now you are ready to design and implement the chatbot. Use the appropriate programming language for the implementation. Once it is implemented successfully, test it to uncover any bugs or errors.

Deep learning based models for chatbots

Generative models are based on deep learning. They are the smartest models for chatbots but are very complicated to build and operate. They give the best response for any query as they use semantic similarity, which identifies the terms that have common characteristics.

The Recurrent Neural Network (RNN) encoder-decoder is the ultimate generative model for chatbots, and consists of two RNNs. As an input, the encoder takes a sentence and processes one word at a time. It translates the series of the words into a predetermined size feature vector. It takes only significant words and removes the unnecessary ones. The encoder consists of a number of hidden layers in which one layer influences the other. The final